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Technical Q&A, S16

Implementation-level answers for `Hunter_SST_second_reference.pptx`, at the level of functions, formulas and thresholds. Backup slides 18 and 19 point here.

Delta convention. mIoU deltas are quoted in points first and as a percentage of the baseline they were measured against second: `+5.17 (+8.1%)` is 5.17 mIoU points, which is 8.1% of the 63.86 it was added to. A percentage is never a share of a total unless the text says so.

Source of every number. `results/s16.zip` for the cells and the two headline rows; the README rows named in `make_s16_figures.py` for the banked levels (63.20, 73.16, 70.30, 71.78, 75.18, 63.86, 69.88, 65.2). The figure script asserts the transcribed constants against the bundle before it draws, so a stale constant fails the build rather than reaching a slide.

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1. What the implicit second reference actually is

`src/occl/adapters/tta.py`.

`rotate_episode(ep)` returns the same episode with every support rotated 180 degrees, image and label map together, via `np.rot90(x, 2)`. The query is never touched: what matters is the support's orientation relative to the query, and rotating the support avoids rotating the prediction back afterwards. Support ids get the suffix `@r180`, because the embedding cache is keyed by the id string and a rotated support served the upright image's features would fail silently.

The pick is `confidence(logits)`:

```
top = logits.max(axis=0)
fg = top > 0.0
return float(top[fg].mean()) if bool(fg.any()) else float("-inf")
```

Mean top-channel margin over the pixels the model claims. A prediction propagated from a mismatched orientation is scattered and weakly positive; one from a matching orientation is compact and strongly positive. An empty prediction returns `-inf` and always loses. No ground truth is consulted, so the choice leaks nothing.

`OrientationTTAAdapter` is a wrapper around the tracker adapter rather than a change to it, so `SSTSam2Adapter` stays byte-stable against every previously recorded number. It counts `n_rotated_chosen / n_total`, which is the fire rate on slide 15.

2. What the explicit second reference actually is

The paper's own multi-shot construction: one 3-frame video {x0, x1, xq}, both supports as conditioning frames, the query last. Selection is memory attention's job, and S6b measured that the bank selects rather than blends.

SAM 2's max_cond_frames_in_attn is -1, meaning all conditioning frames are attended, so both supports are genuinely in play (sam2/modeling/sam2_base.py). This is worth stating because the commissioned SAM 3 checkpoint ships with that same setting configured to 4 (sam3_multiplex_base.py, ++trainer.model.max_cond_frames_in_attn=4), while the SAM 3 tracker's own default is also -1. It is a config attribute rather than weights, so it can be raised, but it is a trap at higher shot counts if nobody looks. It does not arise here.

3. Where the draws come from

seeding.sample_supports and curation.stratified_supports both slice [:shot] from one seeded permutation of the dataset's canonical pool. derive_seed(dataset, group, shot, run_idx) is a blake2b digest that does take a shot field; the nesting comes from its callers, which pass a literal 0 there rather than the actual shot count. sample_supports carries the comment "Do not 'fix' it to shot", because that literal is exactly what makes 1-shot a strict prefix of the 2-shot and 5-shot draws instead of a separate randomisation.

Consequence, and the reason S16 is runnable at all: run i's 1-shot support is the first element of run i's 2-shot draw. The per-seed difference therefore still cancels the shared support's difficulty, which is what makes the row a paired comparison rather than two independent means.

Butterfly used canonical_placement, beetle canonical_balanced, both 20 seeds, 5 groups for butterfly and 1 for beetle, hence 100 and 20 (group, run) pairs.

4. Why registry.compare refuses the row, and what unlocks it

The row crosses protocol.shots and protocol.support_set_hash. Both are comparability fields, so the default refusal is correct: support_set_hash legitimately differs across shot counts, so the field that normally guards pairing cannot stand guard here.

The one sanctioned route is compare(..., nested_shots=True), which calls nested_shot_evidence(treat baseline) before masking anything:

```
def _draws(rec):
    return {(r["group"], int(r["run_idx"])): [str(x) for x in r["support_ids"]]
            for r in (rec.get("runs") or []) if r.get("support_ids") is not None}

bad = [k for k in shared
       if len(dt[k]) != kt or len(db[k]) != kb or dt[k][:kb] != db[k]]
```

It re-reads the per-run support_ids off both records, requires treatment shots > baseline shots, requires at least one shared (group, run), and raises ValueError if the prefix property fails on any pair. Summary-only records are refused outright, because a record with no draw carries nothing to verify.

Only then does it mask NESTED_SHOT_FIELDS = ("protocol.shots", "protocol.support_set_hash"), and the proof travels in the result as nested_shots = {treatment_shots, baseline_shots, runs_checked}. The recorded rows carry runs_checked 100 and 20.

Naming a protocol field in ignore_fields by hand is still refused. The relaxation has to arrive with its evidence. The model.tta toggle the row also crosses is declared separately in ignored_fields, with both values recorded.

The nesting was verified on the real canonical pools before any GPU time was spent: 100 of 100 butterfly and 20 of 20 beetle pairs nested, and every beetle 2-shot draw spans 2 distinct species strata. The gate to combine schema is exercised end to end on synthetic cells in tests/test_s16_gate_roundtrip.py, including the refusal on non-nested draws.

5. The metric, exactly

`METRIC_SPEC = {name: iou_per_trait_macro, assignment: argmax_label, reduction: macro_over_gt_traits}`.

Per query: resize logits to the query's native resolution, argmax across trait channels, background wherever the top channel is not positive, IoU per ground-truth trait, mean over the traits present in that image's ground truth. Then the mean over queries, then over the 20 runs. A trait the model never claims scores 0 and still counts, so a missed antenna costs as much as a botched elytra.

Four readings are recorded per cell and all four are in the bundle:

reading	what it changes	butterfly TTA / 2-shot	beetle TTA / 2-shot
plain	the headline rule above	75.18 / 71.48	63.86 / 69.04
independent	per-trait threshold rather than argmax	75.23 / 71.73	64.94 / 70.27
best-of-3	independent overlap *and* boundary tolerance	76.72 / 73.23	65.45 / 70.75
oracle	best assignment of the proposed regions	75.65 / 75.35	66.93 / 71.73

The second reading is `miou_boundary.independent_best3`, which is what `combine_s16.py` pairs on, so it differs from the plain headline in two ways at once: argmax is replaced by an independent per-trait threshold, and the boundary gets the paper's plus or minus one pixel tolerance. Both readings give the same sign and both are significant, so the conclusion does not depend on either one, but it is a two-factor change and should be described as one.

The oracle reading is recorded per cell and is not paired, so oracle differences quoted anywhere in this deck are differences rather than tested effects.

6. What the oracle ceiling means, and why slide 14 matters

The oracle reading assigns each proposed region to whichever trait scores best. It is therefore an upper bound on what assignment alone could fix, and the gap between plain and oracle is assignment error while the ceiling itself is a property of what the model proposed.

Beetle's ceiling moved 66.93 to 71.73, +4.80 (+7.2%), when the second specimen entered the bank. That is a change in the proposals, not in the assignment. Butterfly's is flat within a third of a point, 75.65 to 75.35. Neither figure carries a paired test, since the oracle reading is not one of the paired metrics.

S14sam3's arch arm moved the same axis on beetle by training, 66.76 to 71.08. Here it moved by data instead. S14sec has no arch arm, so name S14sam3 when making this comparison.

7. The anchors, and what they would have caught

The two 1-shot TTA cells resume the banked S11 ledger cells rather than being re-measured. The evaluator fingerprint is identical, so the run costs zero inference and the values come back bit for bit: every recorded field matches S11 to all 16 significant digits. The gate pins them to 0.7518 and 0.6386 with a tolerance of 0.005, that is 0.5 mIoU points.

That tolerance is written for the case where the cell is genuinely re-run: the plain path is deterministic to about 1e-4, but TTA's confidence pick can flip on near-ties, so 0.5 is generous to noise and still catches the slips that matter. The named failure it guards against is TTA being silently off, which would put roughly 70.30 in a 75.18 cell, and which has happened twice in this project when s5c ran as plain s5b.

What the anchors therefore certify is the wiring of this stage: the right cells in the right slots, drawn from the right pool, merged under the right labels. They are not an independent reproduction of TTA, and the only genuinely new measurements in S16 are the two 2-shot cells, which have no anchor of their own and are guarded instead by the dropped-frame floor in section 8.

8. Disqualifiers written before the run

Gate FAIL, debug before interpreting:

- beetle 2-shot at or below the banked 1-shot plain level 63.20, or butterfly 2-shot below 69.80 (70.30 minus 0.5). A second conditioning frame cannot make the tracker worse unless it is being dropped, so this is the S6 "frames not dropped" check one shot down. It is unpaired against the ledger by design, because the plain 1-shot cell was deliberately not re-run.

Recorded but not gated:

- beetle 2-shot above 73.16, the measured 5-shot level, would mean the draw is wrong, most likely two strata collapsing to one, rather than that 2-shot is unusually good. Measured 69.04, so this stayed quiet.

9. The seven gate predicates, per dataset

1. Both cells on the dataset's canonical protocol, cell names matching their shot counts and TTA flags, 20 of 20 seeds each.
2. The 1-shot TTA anchor reproduces the S11 ledger within 0.5.
3. The headline carries its per-run prefix-nesting proof, with `runs_checked` equal to 20 times the number of groups.
4. The headline declares the `model.tta` toggle it crosses.
5. 2-shot clears the banked 1-shot level, so no conditioning frame is dropped.
6. The paper's best-of-3 ran alongside the plain metric.
7. The headline is recorded with its confidence interval and seed count.

Seven per dataset, 14 of 14 PASS, landed 2026-09-01.

10. Statistics

Paired bootstrap over the 20 seeds, 10,000 resamples, seed 0, computed by `registry.compare`. Significance means the 95% interval excludes 0.

dataset	delta	CI95	seeds improved	paired std
butterfly Major	-3.71	[-4.48, -2.69]	2 of 20	2.10
beetle	+5.17	[+2.63, +7.77]	15 of 20	6.05

Butterfly's 2 of 20 with a tight interval and beetle's 15 of 20 with a wide one are consistent with the cell spreads: butterfly's 2-shot cell has std 0.30 and beetle's has 4.28.

11. Cost

Two `fanout` passes per dataset. Butterfly about 1.1 hours, dominated by the TTA cell's double decode; 2-shot adds one conditioning frame at roughly 1.2 times the plain cost. Beetle about 0.3 hours. Both inside one A100 session, and less when the TTA cells resume from the Drive bank, since an identical evaluator fingerprint means zero inference.

Splittable per dataset via `S16_DATASETS`; the merge pulls a missing dataset's combine from Drive. Gate: `check_gate.py s16` on the merged bundle, or `--bundle s16_<dataset>.zip` for a split session.

12. What S16 does not claim

- It does not measure whether TTA and an explicit second shot compose. Four runs, two cells per dataset, no `{1,2}-shot` by `{plain, TTA}` square. The square was cut to these four runs by decision on 2026-09-01, and the cut single-lever deltas remain reconstructible against S11's banked plain cells.
- It does not say 2-shot beats the trained stack. On beetle 69.04 at two labels is within noise of 69.88 at one label plus training, which is a statement about label cost. Nor is a second label the largest lever measured on beetle in absolute terms: S6's 1 to 5 shot lift is +9.96 on the same stock model.
- The +5.84 and +1.18 quoted against banked 1-shot levels are unpaired cross-stage subtractions, since the plain 1-shot cells were deliberately not re-run. They carry no confidence interval, and the gate says so in the same words.
- It does not generalise across datasets. The two results have opposite signs, each by the mechanism that dataset's residual named.
- It says nothing about butterfly's absolute gap to the paper's Major numbers, which is a separate open item.

Short answers

- **Why is TTA "implicit"?** The rotated support is a second reference that never enters the memory bank and is chosen per episode by a confidence rule instead of by attention.
- **Why does TTA fire on 29.7% of beetle episodes but pay only +0.66?** Firing means the rotated pass was more confident. It adds an orientation, not a species, and beetle's residual is species identity.
- **Why does 8.1% firing pay +4.88 on Major?** The cohort it is presumed to be reaching is cyrbia's rotated queries, 37% of that subspecies' test specimens, which score 0.48 against 74.95 for the upright ones, and which nothing support-side reaches. Only the aggregate fire rate is recorded, so the per-episode correspondence is an inference, not a measurement.
- **Is the butterfly band miss a failure?** It is a miss of 0.19 on the 2-shot side, and the direction and significance were both predicted. The cause is measured: the shot curve saturates.
- **Why 20 seeds?** It is the project's standing protocol, and it is what the banked cells the anchors pin to were run with.
- **Could the two cells differ in anything other than the second reference?** Both are base SAM 2 with no adapters, the same image size, dtype and normalisation, the same canonical pool, the same seeds, and the same evaluator. The comparison records every field it masks.